

# Equation audit of arm revised: four known defects, one new one, and what the fixes do to the calibration

The catalogue's split rows; the 6–18% price jump at the spin-up restart, traced to the money wage (not the Sraffian globals); the Armington origin mismatch between real final demand and final imports, closed as change R10; the `va_g` / `va_index` definitions; and a numerical identity check that finds expenditure GDP and income GDP disagree whenever tariffs are non-zero.

LEEDS\_MODEL — JUST2CE

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**Written:** 2026-09-29 · **Repo at:** `main` @ 8e07250 plus this session's uncommitted engine changes (`model/code/run_model_2026.R`, `model/code/MVP_model_2026.R` and the re-generated `flash_MVP_model_2026.R` and `model/code/arms/*`). **Baselines compared** (all `output/scenarios/arms/revised/runs/`, `gitignored`):

- `baseline_pre_wagecarry_20260929.RDS`: the calibrated revised of 28 September, before any fix;
- `baseline_wagecarry_only_20260929.RDS`: with the restart fix only;
- `baseline.RDS`, also kept as `baseline_wagecarry_r10_uncalibrated_20260929.RDS`: with the restart fix and R10. **This is now the cached revised baseline, and it is not calibrated.**

**Addendum, same day: the maintainer's decisions applied (section 8).** The `va` valuation fix (item 6 below) and chain-linked volume growth for `va_g` / `va_index` (item 7) are implemented as change **R11**. Every identity in section 4 now closes, `yn =`

va included. The bank-reserve question is answered from G&L ch. 10 in section 8. The b\_cb\_s\_Z2 closure moves to the item-4 plan.

## Summary

### 1. Split catalogue rows: fixed.

- *Cause:* on 15 equations one column's note carried an arm addendum the others did not, and the page pivots on the note.
- *Fix:* every column now carries the full note. `tools/melt_equation_tables.py` refuses to build if an equation has more than one key tuple.
- `model/equations/model-equations.html` shows 145 equations in 145 rows.

### 2. Restart price jump: cause found and fixed.

- *Cause:* it is not the four Sraffian globals (the hand-off's hypothesis). It is **the money wage**. Under `price_spec = "production"` the wage is a state, but the workbook types it exogenous, so the spin-up restart did not carry it.
- *Fix:* the restart now carries it.  $p(2)/p(1) = 1$  to  $3e-6$ .

### 3. Armington gap: fixed as R10. Real final demand now uses the same Armington-adjusted origin shares as the import volumes and the deflators. The engine-side mismatch falls from 5.61 to below $3e-9$ .

### 4. The older arms are untouched. submitted, lambda and sraffian re-run identical() to their caches: 0 of 671,300 / 671,900 / 671,900 cells differ.

### 5. Consequence: revised must be recalibrated. The 28 September calibration had absorbed the price jump. Without the jump:

- the price level is 1.004 (EU) and 1.022 (RoW), where it was 1.089 and 1.188;
- nominal value added is 0.95 and 0.85 of target;
- the EU current account runs +6.4 per period, so the baseline is no longer stationary.

### 6. New defect, not fixed: expenditure GDP income GDP when tariffs are non-zero.

- `yn` (eq. 2.4) is below `va` (2.4b) by 13.9 (EU, 1.3% of VA) and 12.3 (RoW, 0.4%).
- With tariffs set to zero the difference is  $4e-6$ .
- Every stock-flow identity still closes, because profits are computed from `yn`. But `va` is the calibration target and the denominator of `debt/VA`.

### 7. va\_g / va\_index: a decision for the maintainer (section 5), not changed.

## 1. The catalogue's split rows

- **What the page does.** `tools/build_equation_tables.sh` pivots `qmd/equations/equations.csv` with the row key `eq_num`, `eq_id`, `block`, `variable_name`, `dependent_variable` plus notes.
- **The cause.** Fifteen equations had more than one distinct note: 0.1, 2.1, 2.4, 3.3, 3.5, 3.6, 5.2, 8.1, 8.3–8.7, 8.sraffian.1 and 9.1c. Equation 8.sraffian.1 also had two dependent variables, `D.matrix` and `K.matrix`.
- **Every case had the same form, checked mechanically.** The shorter note is an exact prefix of the longer one, which adds a dated addendum ("ARM REVISED, R9 ...", "CATALOGUE CORRECTED 2026-09-28 ..."). Giving every column the longest note therefore loses nothing.
- **What changed.** 131 rows of the CSV (10 equations  $\times$  9 columns,  $5 \times 8$ , and the one revised cell of 8.sraffian.1, whose dependent variable now names both matrices).

- **The guard.** `tools/melt_equation_tables.py` now asserts one key tuple per equation, so the split cannot silently return.

## 2. The price jump at the spin-up restart

**Observed (the hand-off's defect 3).** In the calibrated revised baseline  $p(2)/p(1)$  ran from 1.059 to 1.178 across the 108 goods, and prices were flat after  $t = 2$ .

**The hypothesis tested first, and rejected.** The hand-off suspected that the restart did not carry `R_max`, `r_sraf`, `pi_std` and `p_std`, which are all 0 at  $t = 1$ . That is true, but at  $t = 2$  `R_max` and `r_sraf` come back **exactly** at their spin-up values (0.596984 and 0.243886). Their lagged values do not drive prices.

**The measurement.** I ran the 60-period spin-up alone, then the restarted run, and listed every row whose  $t = 1$  value differs from the spin-up's last period. Apart from the ecological accumulators, which restart from the workbook by design, the list contained **all 108 money-wage rows w**.

**The mechanism.**

- **Why w moves.** Under `price_spec = "production"` (arms sraffian and revised) each period replaces the sectoral wages by the region's employment-weighted mean (`MVP_model_2026.R`, eq. 6.9a). So `w` is a state variable that evolves.
- **How far it moved.** Over the spin-up it moved from 0.0333 to 0.0297 in the EU and from 0.0111 to 0.00636 in RoW.
- **Why the restart dropped it.** The carry rule in `run_model_2026.R` carries rows typed `endogenous`, and the workbook types `w` as `exogenous`. The restart therefore put the 54 sectoral workbook wages back at  $t = 1$ .
- **The size of the reset.** At  $t = 2$  their employment-weighted mean was 0.03120 (EU, +5.0% on the spin-up's) and 0.007498 (RoW, +17.9%).
- **Why prices jumped.** The price of production  $p = w \text{ pr} + (1 + r) p (A + K)$  is homogeneous of degree one in `w`. That gives the observed jump, largest in RoW.

**The fix** (`run_model_2026.R`, carry rule). When `price_spec = "production"` the restart also carries `w` and the four Sraffian globals. The code already did the same for `wage_spec = "growth"`, for the same reason.

**The fix measured on a 6-period run after the spin-up:**

- $p(2)/p(1)$  lies in  $[0.999997, 1]$ ;
- $w(2)/w(1) = 1$  exactly;
- `R_max`, `r_sraf`, `pi_std` and `p_std` at  $t = 1$  equal their spin-up values;
- consumption moves  $701.4 \rightarrow 697.7 \rightarrow 695.0$  at  $t = 1, 2, 3$ , where it jumped  $701.4 \rightarrow 725.7 \rightarrow 659.3$  before.

**A residual step in the consumption deflator.** `pa` still steps +0.43% (EU) and +0.15% (RoW) at  $t = 2$ , while `p` does not move.

- *Why:* the Armington shares are computed relative to the prices of  **$t = 1$  of the current run** (`base$p`).
  - In the spin-up that reference is the workbook's prices.
  - In the main run it is the spin-up's end prices.
  - At  $t = 2$  the adjusted shares therefore return to the stored (2011 data) shares.
- *Assessment:* a one-time re-basing, and arguably the right one. In the main run the baseline now buys exactly the data shares at the stationary prices.

- *It also removes most of the baseline Armington gap by itself* (section 3).

**What the fix does to the calibration.** Real activity is on or above target, but the nominal targets, the debt and the external balance moved:

| at t = 100                             | calibrated 28 Sep<br>(with the jump) | restart fix only | restart fix + R10 |
|----------------------------------------|--------------------------------------|------------------|-------------------|
| pa, EU / RoW                           | 1.089 / 1.188                        | 1.007 / 1.017    | 1.004 / 1.022     |
| nominal VA / target,<br>EU / RoW       | 1.000 / 1.001                        | 0.959 / 0.847    | 0.953 / 0.853     |
| real consumption /<br>target, EU / RoW | 1.055 / 1.017                        | 1.098 / 1.005    | 1.092 / 1.007     |
| debt/VA / target, EU<br>/ RoW          | 1.000 / 1.000                        | 0.110 / 1.100    | 0.733 / 0.920     |
| EU current account,<br>mean t = 76–100 | +0.01                                | +7.41            | +6.36             |
| fitness (scale =<br>"gdp")             | 0.00082                              | 0.046            | 0.0083            |

- **Why the old calibration no longer holds.** The enterprise premium `e` was derived (`R/calibrate_enterprise_premium.R`) so that prices of production reproduce the observed price level. Now that prices stay there, the tax scales, the equity share and the debt endowment of the 28 September search are compensating for an artefact that has gone.
- **Recalibration is needed.** It is not done here, because three open decisions (section 7) move the same targets.

### 3. The Armington gap: change R10

**The defect (hand-off defect 2).**

- Real final demand `d` (eq. 2.1) split each region's purchases by origin with the **stored** shares `, , , _g`.
- The final-import volumes (9.1c) and the deflators (8.4–8.6) used the **Armington-adjusted** shares (9.1e, `= 0.626`).
- So the partner output produced for a region's final use and that region's final imports differed.

**The fix, R10, gated to `revision_spec = "revised"`.**

- **The hoist.** The Armington block (the computation of `arm1` and `arm2`) is moved, unchanged, above `d`. All its inputs are predetermined within the period: the stored shares written at the top of the period, lagged `p`, `, VAT` and `xr`, and the `t = 1` reference. So the move is numerically identical for every arm, and the re-runs of `submitted`, `lambda` and `sraffian` confirm it cell for cell.
- **The change.** Under R10, `d` reads `arm1/arm2`. The R9 replacement term subtracts  $\tilde{\times} da$  with the same adjusted `.`

**The check, made non-tautological.** `R/measure_armington_gap.R` computed the “produced” side from the stored shares by construction, so after R10 it would still report the old gap. It now also reports `engine_gap`:

- *the test:* the engine's own `d` for a region's goods, minus (home final use at the adjusted shares + the partner's final imports as the engine computed them);
- *what it must give:* 0 when volumes and imports agree.

Table 2: Baseline. **gap\_stored\_vs\_adjusted**: final imports computed with the stored minus the adjusted shares. **engine\_gap**: the engine’s d for the region’s own goods minus home use plus partner imports; it must be 0.

| run                        | t   | region | gap_stored_vs_adjusted | engine_gap    |
|----------------------------|-----|--------|------------------------|---------------|
| calibrated 28 Sep (no R10) | 2   | Z1     | 0.0000000000           | -0.0000056159 |
| calibrated 28 Sep (no R10) | 2   | Z2     | 0.0000000000           | -0.0000235346 |
| calibrated 28 Sep (no R10) | 70  | Z1     | 2.1262833601           | -5.6057511815 |
| calibrated 28 Sep (no R10) | 70  | Z2     | -3.4794677055          | 5.6057505722  |
| calibrated 28 Sep (no R10) | 100 | Z1     | 2.1232142952           | -5.6053306562 |
| calibrated 28 Sep (no R10) | 100 | Z2     | -3.4821163224          | 5.6053304534  |
| restart fix + R10          | 2   | Z1     | 0.0000000000           | -0.0000001142 |
| restart fix + R10          | 2   | Z2     | 0.0000000000           | -0.0000004567 |
| restart fix + R10          | 70  | Z1     | -0.0000258761          | 0.0000000000  |
| restart fix + R10          | 70  | Z2     | 0.0000407413           | -0.0000000002 |
| restart fix + R10          | 100 | Z1     | -0.0000178767          | -0.0000000007 |
| restart fix + R10          | 100 | Z2     | 0.0000277283           | -0.0000000029 |

- **Before R10** the engine gap is 5.606 for each region’s goods: the EU gap (+2.126) minus the RoW gap (−3.479), because the home-use block moves opposite to imports.
- **After R10** it is −4.6e-7 at t = 2 (within the Gauss–Seidel tolerance) and below 3e-9 at t = 70 and t = 100, which meets the hand-off’s acceptance test of 1e-8.
- **Why the baseline test is weak on its own.** With the restart fixed, the stored and adjusted shares nearly coincide in the baseline (the stored-vs-adjusted gap is about 1e-8), because the reference prices are now the stationary ones. The scenario test below is the one that bites.

**Under scenario 7**, re-run on the R10 engine (output/scenarios/arms/revised/runs/shock\_7\_run.RDS, on the uncalibrated baseline):

- **The pre-R10 mismatch.** At t = 100 the stored-versus-adjusted gap is +0.0092 (EU) and −0.0133 (RoW). That is the mismatch the pre-R10 engine would have produced.
- **Measured against the scenario’s own effect.** RoW’s final imports rise by 0.0175 in the scenario, so the mismatch would have been about 76% of the response.
- **With R10.** The engine gap stays at 2.5e-7 or less: 2.5e-7 at the shock period t = 70, and 2.7e-8 at t = 100. That is the Gauss–Seidel convergence tolerance, not a model gap.
- **Against the acceptance test.** At t = 70 of the scenario the gap is above the 1e-8 threshold, by solver noise only; in the baseline it is below the threshold at t = 70 and t = 100.

#### 4. Numerical identity check

R/check\_identities.R (new) evaluates the accounting identities on a cached run, period by period. Each identity closes two *different* equations against each other, so it tests something rather than restating one line of code. The table shows the largest |residual| over t = 3–100.

Table 3: Largest absolute residual, t = 3–100, in 10 bn EUR.

| eq. | identity                                                                                                            | region | calibrated   | restart fix  |              |
|-----|---------------------------------------------------------------------------------------------------------------------|--------|--------------|--------------|--------------|
|     |                                                                                                                     |        | 28 Sep       | + R10        | + R11        |
| 1.5 | household wealth: $V = V(-1) + YD - c \text{ pA}$                                                                   | Z1     | 1.479000e-09 | 5.300000e-11 | 9.300000e-11 |
| 7.7 | household balance sheet: $V + Lh = Hh + Mh + B(\text{own}) + B(\text{foreign}) + E(\text{own}) + E(\text{foreign})$ | Z1     | 2.400720e-07 | 5.713000e-09 | 1.004400e-08 |

| eq.         | identity                                                                                                               | region | calibrated   | restart fix  |              |
|-------------|------------------------------------------------------------------------------------------------------------------------|--------|--------------|--------------|--------------|
|             |                                                                                                                        |        | 28 Sep       | + R10        | + R11        |
| 4.1         | loans: $L_s = L_F + L_h$                                                                                               | Z1     | 3.350030e-07 | 8.046000e-09 | 1.414000e-08 |
| 4.4         | bank balance sheet: $L_s + B_b = M_s + A$                                                                              | Z1     | 0.000000e+00 | 0.000000e+00 | 0.000000e+00 |
| 4.2         | deposits: $M_s = M_h$                                                                                                  | Z1     | 1.764020e-07 | 4.216000e-09 | 7.419000e-09 |
| 5.6a        | debt: $B_s = B_s(-1) + DEF$                                                                                            | Z1     | 0.000000e+00 | 0.000000e+00 | 0.000000e+00 |
| 10.2/10.3   | bill market: $B_s = \text{held by own hh} + \text{foreign hh} + \text{banks} + \text{CBs}$                             | Z1     | 0.000000e+00 | 0.000000e+00 | 0.000000e+00 |
| 3.4         | capital: $k = k(-1) + id - da$                                                                                         | Z1     | 0.000000e+00 | 0.000000e+00 | 0.000000e+00 |
| 2.4 vs 2.4b | GDP: expenditure $Y_N = \text{income } VA$                                                                             | Z1     | 9.893001e+00 | 0.385937e+09 | 0.184400e-08 |
| 9.8         | NAFA = DEF + CAB (as coded)                                                                                            | Z1     | 0.000000e+00 | 0.000000e+00 | 0.000000e+00 |
| 1.5         | household wealth: $V = V(-1) + YD - c \text{ pA}$                                                                      | Z2     | 1.535000e-08 | 5.350000e-10 | 9.330000e-10 |
| 7.7         | household balance sheet: $V + L_h = H_h + M_h + B(\text{own}) + B(\text{foreign}) + E(\text{own}) + E(\text{foreign})$ | Z2     | 4.696100e-08 | 1.304000e-09 | 2.286000e-09 |
| 4.1         | loans: $L_s = L_F + L_h$                                                                                               | Z2     | 2.474000e-07 | 6.999000e-09 | 1.221700e-08 |
| 4.4         | bank balance sheet: $L_s + B_b = M_s + A$                                                                              | Z2     | 0.000000e+00 | 0.000000e+00 | 0.000000e+00 |
| 4.2         | deposits: $M_s = M_h$                                                                                                  | Z2     | 9.214070e-07 | 2.625600e-08 | 4.584100e-08 |
| 5.6a        | debt: $B_s = B_s(-1) + DEF$                                                                                            | Z2     | 0.000000e+00 | 0.000000e+00 | 0.000000e+00 |
| 10.2/10.3   | bill market: $B_s = \text{held by own hh} + \text{foreign hh} + \text{banks} + \text{CBs}$                             | Z2     | 0.000000e+00 | 0.000000e+00 | 0.000000e+00 |
| 3.4         | capital: $k = k(-1) + id - da$                                                                                         | Z2     | 0.000000e+00 | 0.000000e+00 | 0.000000e+00 |
| 2.4 vs 2.4b | GDP: expenditure $Y_N = \text{income } VA$                                                                             | Z2     | 2.014309e+01 | 0.229886e+03 | 0.466730e-07 |
| 9.8         | NAFA = DEF + CAB (as coded)                                                                                            | Z2     | 0.000000e+00 | 0.000000e+00 | 0.000000e+00 |
| 9.3         | world trade: $\text{exports}(Z1) = \text{imports}(Z2)$ , real                                                          | world  | 0.000000e+00 | 0.000000e+00 | 0.000000e+00 |
| 9.5         | world trade balance: $TB(Z1) + xr \text{ } TB(Z2) = 0$                                                                 | world  | 0.000000e+00 | 0.000000e+00 | 0.000000e+00 |
| 9.6         | world current account: $CAB(Z1) + xr \text{ } CAB(Z2) = 0$                                                             | world  | 0.000000e+00 | 0.000000e+00 | 0.000000e+00 |
| 10.6        | world gold: $dOR(Z1) \text{ pOR} + dOR(Z2) \text{ pOR} = 0$                                                            | world  | 2.521349e-05 | 6.565000e-07 | 1.166201e-06 |
| run.model   | run.model's own consistency error (last iterate)                                                                       | world  | 3.180000e-10 | 0.000000e+00 | 0.000000e-12 |

**Every stock–flow identity closes to 1e-6 or better:** the household wealth and balance sheet, the bank balance sheet, loans, deposits, the debt law of motion, both bill markets, capital, world trade, the world current account and world gold. **One valuation identity does not.**

#### The new defect: expenditure GDP – income GDP when tariffs are non-zero

- **The size.**  $y_n$  (2.4, expenditure side, net of VAT and tariffs) is below  $v_a$  (2.4b, income side,  $\sum p_x - (p_A + xr) x$ ):
  - by 9.9 (EU) and 20.1 (RoW) in the calibrated run;
  - by 13.9 and 12.3 in the fixed one;
  - by about  $-12 / -20$  already at  $t = 1$ .

- **The test.** A 4-period run without the spin-up, varying the tariff pass-through (the share of the duty borne by the importer) and the tariffs:

| run               | EU $yn - va$ , $t = 2-4$ | RoW $yn - va$ , $t = 2-4$ |
|-------------------|--------------------------|---------------------------|
| $= 0.5$ (shipped) | -38.7, -13.0, -13.2      | -29.8, -17.7, -13.1       |
| $= 1$             | -20.9, -16.6, -12.3      | -31.4, -4.8, -7.7         |
| $= 0$             | -56.6, -0.1, -11.4       | -27.9, -54.8, -26.3       |
| all tariffs = 0   | 0.0000                   | 0.0000                    |

(The first periods without a spin-up are unsettled, hence the noise.)

- **The conclusion.** The gap exists if and only if tariffs are non-zero, and no value of  $\alpha$  closes it.
- **The mechanism, hypothesised, not yet decomposed numerically.** Two valuation terms, both making  $va$  too high:
  1. **exports:** with  $\alpha < 1$  the exporter receives  $CV = xr \cdot p(1 + \alpha)/(1 + \alpha) < p$  per unit exported, but  $va$  counts the full  $p$ ;
  2. **imported intermediates:**  $va$  charges them at the exporter's price  $p$  (the  $p A_{xr}$  term), not at the duty-inclusive price the importing firm pays, which  $yn$  (through IMP and TAR) does charge.

The second term does not vanish at  $\alpha = 1$ , which fits the table.

- **Why the stock identities still close.** Firms' profits  $FF$  are computed from  $yn$ , so every flow reaching a balance sheet is consistent. What is mis-measured is the statistic  $va$ . But  $va$  is also:
  - the calibration target;
  - the denominator of  $debt/VA$ ;
  - the base of  $va_j \rightarrow int_j$  (1.8) and the sectoral distribution variables.
- **Proposed fix, for a decision.** Under `revised`, compute  $va_j$  as revenue actually received minus intermediate inputs at the buyer prices  $q_a$ , so that  $\sum va_j = yn$ . It moves the VA target and  $debt/VA$ , so it belongs in the same recalibration.

## 5. $va_g$ and $va_{index}$ : a definition to choose

- **rva and va** (2.4c, 2.4a).  $rva$  is current volumes at *last period's* prices, and  $va$  is current volumes at current prices.
- **So  $va_g$**  (2.4e)  $= rva/va - 1 = 1/P - 1$ , where  $P$  is the Paasche price index of value added. That is approximately **minus the inflation rate**.
- **The maintainer's "1 - RVA/VA"** is approximately **plus** the inflation rate.
- **Neither is volume growth.** The standard chain-linked volume growth (Laspeyres at previous-period prices) is  $rva_t / va_{t-1} - 1$ . The matching index is  $va_{index_t} = va_{index_{t-1}} \cdot rva_t / va_{t-1}$ , starting at 1.
- **$va_{index}$  is 0 in every period** because it chains from its workbook value 0.
- **Nothing in the engine reads either variable**, so either choice changes no other variable. A change to them in `submitted` would still break that arm's bit-for-bit contract, so it would be gated to `revised`.
- **Not changed; waiting on the choice.**

## 6. Other findings

- **Bank reserves earn nothing** (qmd/reports/2026-09-29-calibration-ratios.qmd).
  - Bank profit (4.5) has no term in A.
  - Under R7, EU banks hold 59% of deposits as zero-interest reserves and RoW banks 49%. Their profit is negative in every period.
  - The deposit rate (0.0226 EU) is also above the bill rate (0.0216).
- **The EU central bank's RoW-bill stock is frozen at 0** (R6, 10.5).
  - G&L's OPENFIX makes exactly this stock the moving reserve, with gold exogenous (qmd/reports/2026-09-29-reserve-rule-and-lambda-options.qmd).
  - R6 instead settles in gold and freezes the stock at its  $t = 1$  value, 0.
  - A closure choice, carried to item 4 of the hand-off.
- **The durable-goods stock  $dc$  (12.5)** is built from the stored , not the adjusted ones. It is an ecological accumulator, left as is and recorded here.
- **The markup rows  $\mu$**  are not carried across the restart either. They are inert for prices:  $p$  is continuous across the restart after the fix.

## 7. What is left

1. **Decisions, all of which move the calibration targets, so one recalibration covers all:**
  - the  $va$  valuation fix (section 4);
  - the bank-reserve fix;
  - the  $b\_cb\_s\_Z2$  closure;
  - the  $va\_g$  definition.
2. **Recalibrate revised** in the order of 28 September:
  - the equity share  $Z1\_lambdaE0\_2$ , for EU  $CAB = 0$  (R/calibrate\_revised\_cab.R);
  - the tax scales, for the VA level, and the debt endowment (R/calibrate\_revised\_debt.R). The pre-fix calibrated baseline is kept as `baseline_pre_wagecarry_20260929.RDS` for comparison.
3. **What this audit did not do.** It did not walk the text of all 145 revised equations against the code. It tested the accounting numerically instead: section 4 covers stock–flow consistency and one valuation identity. A text-versus-code walk of the price and trade blocks (8.x, 9.x) is still open.

## 8. Addendum: the decisions applied

**R11: value added at the prices actually received and paid, and chain-linked volume growth**

The new definition, under `revision_spec = "revised"`:

$$va_j = p_j x_j - (p_j - x r^f CV_j^f) m_j^f - \mathbf{q}_a^\top A_{.j} x_j.$$

- $m^f_j$  is the partner's imports of good  $j$ : intermediate, final and replacement (`import_volumes()`).
- $CV^f_j$  is what the exporter receives per unit.
- $\mathbf{q}_a$  is the firm buyer-price vector: CIF, duty included, on partner goods.
- $rva_j$  is the same expression at the previous period's prices, on this period's volumes.

- So  $va\_g = rva\_t / va_{t-1} - 1$  is chain-linked (Laspeyres) volume growth, and  $va\_index$  is based at 1 at  $t = 1$  of the main run.

**Where it sits in the engine.** It is computed in the late consistent block, where the import volumes and customs values exist. The two sector readers  $t\_j$  and  $int\_j$  are re-solved there, because the early coded lines run ahead of them on every Gauss–Seidel sweep.

**Measured on the re-cut, still uncalibrated, baseline:**

- $yn - va$  falls from 13.9 (EU) and 12.3 (RoW) to  $9.2e-8$  and  $3.5e-7$ ;
- $\Sigma va\_j = va$  exactly;
- VA falls by 1.3% (EU) and 0.3% (RoW);
- the debt stocks and the deficits are unchanged to four digits, because they were built on  $yn$  already;
- $va\_index$  ends at 0.992 (EU) and 1.025 (RoW), the drift of an uncalibrated baseline;
- `submitted`, `lambda` and `sraffian` are still `identical()` to their caches.

**Bank reserves: the G&L option and the realistic option**

**What the engine does now (R7):**

- banks hold bills as a fixed 10% of deposits;
- advances are the residual, so a negative value is reserves;
- reserves earn nothing;
- the deposit rate is above the bill rate in the EU.

**What G&L do in Model INSOUT, ch. 10, as read in the book:**

- *Required reserves.* Banks hold a **required** reserve  $\cdot M$  in high-powered money (eq. 10.66, p. 335), which earns nothing.
- *Bills are the residual.* **Bills are the banks' residual buffer**,  $B\_b = M - L - H\_b$  (10.67, 10.71, p. 337): “in the case of the banking system, Treasury bills play the role of a buffer” (p. 336).
- *Advances only as a backstop.* **Advances are only a backstop and are never negative.** Banks borrow only when their net bill holding would fall below a floor  $bot \cdot M$  (10.69–10.70). Advances pay  $r\_a = r\_b$  (10.60), and that interest enters both central-bank profit (10.61) and bank profit (10.77).
- *The rate hierarchy.* G&L impose  $r\_l > r\_b > r\_m$ , following Godley and Cripps (1983) (p. 339).
- *The deposit rate is the equilibrating instrument.* It moves so as to keep the net liquidity ratio  $B\_b/M$  between  $bot$  and  $top$  (10.73–10.76, p. 338). When banks hold too many bills,  $r\_m$  falls and households move out of deposits.

**Why the G&L option cannot be switched on in LEEDS today.**

- *Deposits are too large.* They are 77% (EU) and 97% (RoW) of household net wealth. They are the portfolio residual (7.7), and every behavioural is 0.
- *So the residual bill holding exceeds the bill supply.* If bills were the banks' residual, EU banks would need  $M - L = 1,122$  of bills out of a supply of 875. The EU central bank's own-bill holding would be  $-548$  (RoW:  $-2,673$ ).
- *The mechanism that would shrink deposits is disabled.* G&L's equilibrating mechanism, a deposit rate that moves portfolios, needs non-zero rate sensitivities, and all of them are 0.

**Correction to my earlier suggestion.** In the calibration-ratios report I suggested raising `bank_bill_ratio` to about 0.69 (EU) and 0.59 (RoW) so that advances are near 0. That is **infeasible** for the same reason: the banks would demand more bills than the government has issued.

**The realistic option.** Central banks have run floor systems since 2008.

- *The system.* Reserves are remunerated at or near the policy rate.
- *Sizes (recalled, not checked against ECB data).* The euro area's excess reserves were a few per cent of deposits, not tens of per cent.

The realistic option therefore has two parts:

1. **Size:** a household portfolio that holds a realistic deposit share. That is the design of the item-4 plan: home bias and deposit shares from data.
2. **Price:** pay interest on reserves and on advances, symmetrically, at the policy rate  $r^*$ , in both bank and central-bank profit. G&L count interest on advances in both profits; LEEDS omits it.

**Why remunerating today's stock alone would mislead.** On today's stock, remuneration alone would move about  $r^* \times 959 = 11.6$  per period from the EU central bank to the banks:  $r^* = 0.0121$ , on 959 of reserves. The central bank's profit (8.9 today) would turn negative, and the transfer would appear as about 1.1% of GDP of extra deficit. That would turn an unrealistic stock into an unrealistic fiscal cost.

**Recommendation.**

- Decide the reserve regime together with the block in the item-4 plan, because the stock is a portfolio artefact.
- The interest term on A should be added in either regime, since both G&L and a floor system have it. It is small once the stock is realistic.

## 9. The re-calibration of revised

**Why it was needed.** The 28 September calibration had absorbed the spurious restart price jump (section 2). After the fixes, **revised** sat off its targets: nominal VA was 0.94 / 0.85 and the EU current account ran +6.4 per period.

**The procedure,** in the same order as 28 September. The runs are in `output/calibration/revised_joint_search` rounds 1, 3, 4, 5, 6 and 7, with round 2 lost to the failure below.

1. **Tax scales, for value added.** A scale multiplies the net income-tax rates `_w` and `_c`.
  - Measured elasticity of VA to the scale: about  $-0.5$  (EU) and  $-0.42$  (RoW).
  - Adopted: 0.91 (EU) and 0.66 (RoW) on the 28 September scales.
  - So the net rates are now the observed rates  $\times 0.6756$  (EU) and  $\times 0.3673$  (RoW).
2. **The EU equity share `lambdaE0_2`, for a zero EU current account.**
  - Measured slope: about 1,050 of current account per unit of the share.
  - Adopted: 0.02064 (was 0.03227).
3. **The debt endowment, for debt/VA.**
  - Adopted: 1,143.0 (EU) and  $-5,243.8$  (RoW).
  - Measured to move debt one for one and nothing else (rounds  $6 \rightarrow 7$ ).

**The values sit in the workbook**, written by `tools/add_revised_parameters.R`. A re-cut driven by the workbook alone reproduces round 7, and `lambda` and `sraffian` are still `identical()` to their caches.

**Two numerical obstacles, both in the first spin-up periods (every stock zero at  $t = 1$ ):**

- **Gauss–Seidel divergence.**

- *The failure:* at the lower tax rates  $t = 2$  converged only at a contraction of about 0.93 per sweep, and  $t = 3$  diverged (error  $1.9e16$  within 25 sweeps).
- *The fix:* under-relaxation, `gs_relax` = 0.7, a new global parameter, 1 (off) for the older arms. Each sweep becomes  $s \leftarrow \gamma f(s) + (1 - \gamma) s$ .
- *Its effect on the solution:* a converged period is the same solution for any  $\gamma$ . But debt is path-dependent (R6), and the damped iteration reaches a different fixed point in the pathological first periods. At the same parameters, activity matched to 4 digits while pre-endowment debt/VA moved from 1.17 / 0.62 to 2.46 / 0.24. The endowment was therefore calibrated at  $\gamma = 0.7$ .
- *The cap:* `max.iterations` was raised to 1,000, because the damped first period needs about 900 sweeps.

- **$0 \times \text{Inf}$ .**

- *The failure:* at  $t = 2$  of the spin-up a region's equity stock could be exactly 0, making the return on equity  $r_e = (1 - \gamma)FF/e_s$  infinite. The portfolio equations multiply the lagged  $r_e$  by  $\gamma$  coefficients that are 0, so every financial stock became NaN at  $t = 3$ .
- *The fix:* a non-finite lagged  $r_e$  is now read as 0.
- *Why no other arm moves:* the guard is inert whenever  $r_e$  is finite, and any run that had met the infinite value would have crashed.

**The result at  $t = 100$** , from `output/calibration/calibration_ratio_cells_20260929_recalibrated.csv`:

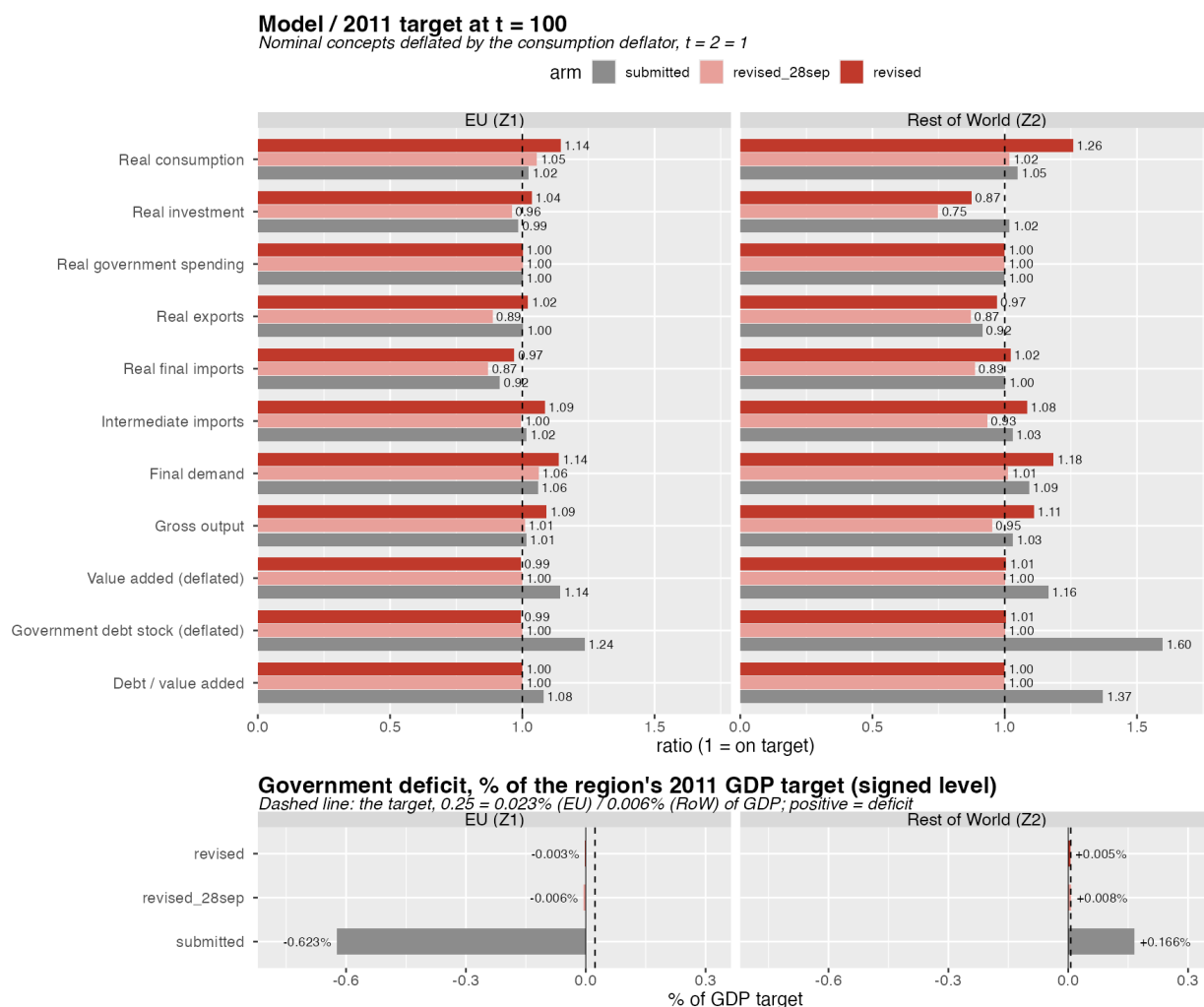


Figure 1: Model / 2011 target at t = 100: submitted, revised as calibrated on 28 September (with the restart artefact), and revised re-calibrated.

#### On target:

- value added: 0.994 (EU) and 1.005 (RoW);
- debt/VA: 0.8105 and 0.8499;
- EU current account: +0.002 per period.

#### Closer to the data than on 28 September:

- exports and imports: 0.97–1.02, from 0.87–0.89;
- RoW investment: 0.87, from 0.75.

#### Further from target:

- real consumption: 1.14 (EU) and 1.26 (RoW), from 1.05 / 1.02;
- final demand: 1.14 / 1.18;
- gross output: 1.09 / 1.11.
- The fitness is 0.0074, against 0.00082 on 28 September. The old figure was flattered by the price artefact, which let nominal VA reach target with lower real volumes.

**The trade-off.** At the price level the model now reproduces ( 1), hitting nominal VA needs lower tax rates, and without transfers most of the extra disposable income is consumed. The model's

consumption share of VA ( $c \cdot p\_A/VA$ ) is 0.67 (EU) and 0.77 (RoW), against 0.58 and 0.60 in the 2011 data.

- *The RoW net income-tax rates are 8.5% (wages) and 6.3% (capital income).*
- *The alternative:* target real consumption instead. VA would then sit below its target.
- *The structural fix:* model transfers, so that gross tax rates can stay near the observed ones.

**How settled the baseline is.** It is stationary in the limit, but still settling after the restart:

- the RoW deficit decays from 25.5 at  $t = 2$  to 0.83 at  $t = 70$  (the shock date) and 0.20 at  $t = 100$ ;
- RoW VA rises about 3% over the run.

A longer spin-up would settle it before  $t = 1$ . It would also move the path-dependent debt level, so the endowment would need re-setting.

**Verification run:** `submitted, lambda, sraffian` re-run on the regenerated flash engine, `identical()` to their caches. `Rscript R/verify_arm_fold.R --periods 12:` all nine arms PASS, folded == switched on every cell (12-period smoke check).